ZhuLong: An Execution-Grounded LLM Agent for EDA Scripting

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Abstract

EDA scripting with tool-specific, often undocumented APIs remains a long-tail bottleneck. We present **ZhuLong**, an execution-grounded LLM coding agent for PyAether, SKILL, and Tcl that combines API retrieval, documentation inspection, and sandbox execution via unified MCP tools into a closed generate–execute–refine loop. We formalize the "grounding gap": EDA correctness is a *situated* property that cannot be guaranteed from static knowledge, bounded by observation fidelity—not model scale.

On **EDA-Eval-PyAether** (158 real-world tasks), the complete system achieves **81.4%** Pass@1 (±1.5), outperforming a pure LLM baseline (**10.5%**). Ablation shows the execution loop is decisive: adding sandbox execution to retrieval yields +9.2 pp, while removing it (−9.2 pp) or retrieval (−0.4 pp) reduces accuracy—consistent with the theoretical prediction. A cross-language reproduction on the SKILL and Tcl slices of **EDA-Scripting-Bench** (449 tasks total) reproduces the same structure—the complete system is the top and the prior-only baseline the bottom on every language.

CCS Concepts

• **Hardware** → **Software tools for EDA**.

Keywords

LLM Agents for EDA, EDA Scripting, Code Generation, Sandbox Execution, Execution Grounding

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1 Introduction

Large language models (LLMs) have demonstrated strong capabilities in general-purpose code generation [17, 23, 25]. However, applying LLMs to EDA scripting—where engineers manipulate in-memory design databases, create layout instances, route schematics, and query netlists—remains challenging: EDA APIs are long-tailed and tool-specific (many rarely appear in public training corpora); correctness cannot be verified statically (it depends on observable side effects in a real EDA environment); and without execution feedback, static generation cannot diagnose errors or iteratively refine [3, 5]. Existing LLM-based EDA work has mainly focused on HDL generation [15, 21], while scripting with real tool APIs—especially with incomplete documentation and sandbox execution—remains underexplored.

This paper presents **ZhuLong**, an execution-grounded LLM coding agent for PyAether, SKILL, and Tcl. Built on Cline-CLI [7], ZhuLong extends three MCP tools [16]—`search_apis`, `get_api_details`, and `run_code`—that form a closed loop: retrieve candidate APIs, inspect their documentation, execute code in the EDA environment, and iteratively refine based on observed errors. All tools accept a unified `lang` parameter, so the same agent logic covers all three languages. We further provide a **theoretical foundation** (Section 4): EDA scripting correctness is a *situated* property that cannot be guaranteed from static knowledge, and the resulting "grounding gap" is bounded by observation fidelity—not model scale.

The main contributions are:

- **An account of *which* factor sets the ceiling, with measurements that distinguish them (RQ1).** We formalize the *grounding gap*: EDA correctness is a situated (L4) property that no offline knowledge can guarantee, bounded by *observation fidelity* not model capacity (Section 4). We answer RQ1 from both sides—laddering Ω with the LLM fixed, and changing $\mathcal{K}$ with the harness fixed—stated *before* the measurements as predictions the experiments could refute.
- **A harness whose necessary components we isolate (RQ2).** ZhuLong couples three MCP tools—`search_apis`, `get_api_details`, `run_code`—so that retrieval (observation), sandbox execution (effector), and the generate–execute–refine loop (belief update) form one closed loop, driven by a single `lang` parameter covering PyAether, SKILL, and Tcl (Section 3). ZhuLong attains **81.4%** Pass@1 on the 158-task PyAether slice, versus **10.5%** (pure LLM) and **72.2%** (RAG). Ablation shows the components are jointly necessary and not interchangeable: adding execution to RAG yields **+9.2 pp**, removing it (−9.2 pp) or retrieval (−0.4 pp) reduces accuracy, and bounding Φ degrades it monotonically.
- **A cross-language benchmark on which the same structure can be re-measured per toolchain (RQ3). EDA-Scripting-Bench** places PyAether, SKILL and Tcl under one task schema, one assertion criterion and one harness (Section 5). The completed PyAether slice (**EDA-Eval-PyAether**, 158 expert-reviewed tasks) is the first standardized PyAether benchmark. We release the harness, task schema and evaluation protocol; task cases derive from vendor documentation and customer designs and cannot be redistributed.

2 Background and Motivation

We target two widely-used EDA scripting interfaces: PyAether for Empyrean Aether [9] and SKILL for Cadence Virtuoso [20]. EDA scripting poses three challenges beyond what generic LLM code generators address: (i) APIs are long-tailed, tool-specific, and poorly documented; (ii) correctness depends on

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side effects in a live, stateful database, not return values; and (iii) no benchmark exists to evaluate whether generated scripts actually work.

Prior work stops short of execution. LayoutCopilot [14], ChaTCL [19], and RAG-EDA [18] combine LLMs with static knowledge bases, but none executes generated code: correctness is assessed by human judgment or retrieval quality, not by whether the script actually modifies a layout. Recent concurrent benchmarks for Tcl/Innovus flows [13, 24] confirm the growing recognition of this gap, yet target different toolchains; a dedicated PyAether/SKILL benchmark remains absent.

Our approach. ZhuLong closes the loop from generation to execution to refinement via sandbox feedback, with a **theoretical grounding** (Section 4) explaining *why* execution is necessary, and **EDA-Eval-PyAether**, the first standardized PyAether benchmark. The contribution boundary is the *harness*—the three MCP tools (`search_apis`, `get_api_details`, `run_code`), the evaluation protocol, and the benchmark; the agent backbone is outside the scope of this paper.

3 System Design

3.1 Overall Architecture

ZhuLong is built on Cline-CLI [7], extended with EDA-specific retrieval, documentation, and execution capabilities. We call the resulting runtime an **ICL-type autonomous agent**: it decides its own next action, keeps no explicit posterior, and maintains its belief state implicitly by in-context learning (formalized in Section 4). As shown in Figure 1, the system comprises three components—the LLM-based agent runtime, the API knowledge base, and the EDA execution environment—interacting through three MCP tools.

3.2 Three MCP Tools

ZhuLong extends three MCP tools [1], each accepting a unified `lang` parameter (`pyaether`, `skill`, or `tcl`) to support all three platforms with identical agent logic:

- `search_apis(query, lang)` retrieves relevant APIs via semantic and keyword-based search. We index APIs by name and description using BGE-M3 embeddings [4] with FAISS [8] for efficient similarity search, returning the top-K results ($K = 5$ by default) with confidence scores and descriptions.
- `get_api_details(api_name, lang)` fetches comprehensive documentation for a specified API, including parameter types, return values, and examples.
- `run_code(code, mode, lang)` executes the generated code in the EDA environment. It supports two modes: (1) **lightweight**—runs in a sandboxed environment without a GUI session (standalone PythonE for PyAether, or SKILL in batch mode for Virtuoso), capturing outputs, errors, and state changes for iterative refinement; (2) **interactive**—runs inside an open EDA session via the agent-EDA bridge (socket communication), enabling operations on unsaved layouts and schematics.

3.3 Agent Workflow

The agent iteratively refines its output through execution feedback [7]. A typical trajectory reasons over the user instruction, retrieves APIs via `search_apis`, inspects candidates via `get_api_details`, generates code, executes it via `run_code`, observes the result, and re-plans—revising code, retrieving alternatives, or terminating—until success or further attempts are deemed unlikely. The agent is not constrained to a fixed sequence: it may skip retrieval when it already knows the required API, or conduct multiple retrieval rounds when initial results lack confidence.

4 Theoretical Grounding: The Grounding Gap

Our empirical results show that the execution loop is decisive: adding sandbox execution to retrieval yields +9.2 pp ($p<10^{-8}$), while removing execution (−9.2 pp) or retrieval (−0.4 pp, n.s.) reduces accuracy. This section explains *why* this is structural rather than implementation-specific: EDA scripting correctness is a *situated* property that cannot be guaranteed from offline knowledge, and execution is the only channel that observes the current design state. The gap is bounded by observation fidelity, not model scale.

4.1 A Four-Level Hierarchy of Information

To make precise what can be learned offline versus what must be observed online, we distinguish four levels of information:

- **L1–Syntactic Correctness**: well-formedness of symbols; acquirable offline.
- **L2–Physical Plausibility**: robust statistical patterns of how the world behaves; acquirable offline as the agent's *prior*.
- **L3–Historical Factuality**: what has occurred or is documented; past-oriented, acquirable offline, but with no connection to what is true *now*.
- **L4–Situated Truthfulness**: what is true *right now* in this specific environment—*indexical* (truth depends on context and changes over time), *non-guaranteeable from offline data*, and requiring *real-time commitment*. **Not learnable offline.**

L4 is not a semantic category but a *grounding requirement* imposed by indexical reference; the four levels separate what can be *known in advance* from what must be *observed now*—L4 is read from the environment, not recalled from memory.

For EDA scripting, the situated state is the live design database: which cell and view are selected, what the layout contains at this moment. The correctness criterion—generated code executes without error and all assertions over the mutated design pass—is an L4 property: a script that is well-formed (L1), plausible (L2), and invokes documented functions (L3) can nonetheless violate an undocumented binding contract or depend on an internal selection state. Such failures must be *observed* by acting on the live design and reading back the result.

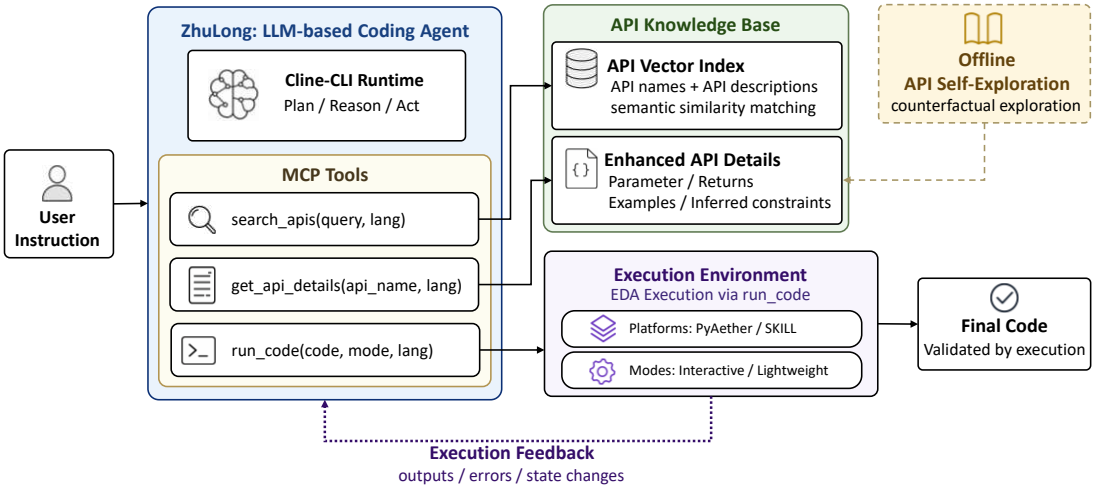


Figure 1: ZhuLong system architecture. The agent interacts with API retrieval, documentation, and EDA execution via three unified MCP tools. Execution feedback supports iterative refinement.

4.2 A Bayesian Formalization of the Grounding Gap

We cast grounding as decision-making under partial observability. Let Θ denote the space of L4 states, o_t a runtime observation, a_t a code action, and $\mathcal{K}$ the offline knowledge in the LLM's frozen parameters (L1–L3). The agent maintains a belief $b_t(\theta) \triangleq p(\theta_t \mid o_{1:t}, a_{1:t-1}, \mathcal{K})$, updated by Bayes' rule:

$$b_t(\theta) \propto \underbrace{p(o_t \mid \theta_t)}_{\Omega} \int_{\Theta} \underbrace{p(\theta_t \mid \theta_{t-1}, a_{t-1})}_{\mathcal{T}} b_{t-1}(\theta_{t-1})\, d\theta_{t-1}, \quad (1)$$

where Ω is the *observation likelihood* (the Harness's observation modules) and $\mathcal{T}$ the *transition model* (the LLM's prior over how EDA operations transform the design). Actions minimize expected cost $a_t^* = \arg\min_a \mathbb{E}_{\theta_t \sim b_t}[\mathcal{C}(\theta_t, a, G) + \gamma V(b_{t+1})]$.

The *grounding gap* is the divergence between the posterior with runtime observations and the posterior from the offline prior alone:

$$D_{\text{grounding}}(t) \triangleq D_{\text{KL}}(p(\cdot \mid o_{1:t}, \mathcal{K}, \text{env}) \,\|\, p_{\mathcal{K}}(\cdot \mid o_{1:t}))\,. \quad (2)$$

This divergence is bounded by observation fidelity, not model capacity. If $I(\Theta_t; O_t \mid \mathcal{K}) = 0$—the observation channel is silent—no prior improvement can recover the current state. Scaling $\mathcal{K}$ improves L1–L3 but cannot push the gap below the floor imposed by Ω. The source is causal: parameters are fixed before deployment, and the environment at time t is not a function of the training corpus. This separates grounding from learning-based adaptation: fine-tuning improves the prior over L1–L3 during *training*; grounding requires the likelihood and a posterior update from observing o_t during *deployment*.

4.3 ZhuLong as an Instance of the Agent Harness

The formulation maps onto ZhuLong's architecture: the frozen LLM supplies $\mathcal{K}$ (L1–L3), $\mathcal{T}$, and Π; the Harness supplies observation, belief update, and execution. `search_apis`/`get_api_details` are the observation modules Ω; `run_code` is the effector acting on the live design database; the re-planning loop implements Φ. The closed loop $b_t \to a_t \to o_{t+1} \to b_{t+1}$ turns one-shot generation into grounded search.

4.4 Design Principles

Three principles follow: *freeze* $\mathcal{K}$, *maximize* Ω's fidelity, and *run* Φ *continuously*. (1) Decouple knowledge from belief: $\mathcal{K}$ is the static L1–L3 prior in the LLM; b_t is the dynamic posterior maintained by the Harness. Freezing $\mathcal{K}$ and absorbing adaptation in b_t yields stability and flexibility. (2) Optimize observation over the prior: the gap is bounded by Ω's fidelity, not model scale; observation investment has direct bounded effect, while model scaling has diminishing returns once the prior is rich. (3) Design for continuous belief updating: Φ's update rate must exceed L4 dynamics, and in an ICL-type autonomous agent the quantity to bound is how much evidence b_t may accumulate—a graded execution budget or a lagged read-back.

These principles *account for* our ablation findings: removing Ω (w/o Sandbox) makes L4 violations unobservable (-9.2 pp); RAG improves the L1–L3 prior but cannot observe L4 (ceiling 72.2%). Three diagnostic questions follow for any candidate system: (1) does it implement Ω with sufficient fidelity? (2) does it maintain b_t through Φ? (3) does it execute a_t through a causal interface? These organize the evaluation into RQ1–RQ3 (Section 6.1).

5 Benchmark Construction

EDA-Scripting-Bench spans three EDA scripting languages—PyAether, SKILL, and Tcl—under one task schema and one execution-based protocol. Every task carries a `lang` field and language-appropriate assertions, so a single harness (Section 3.3) can be scored uniformly across platforms. It comprises **EDA-Eval-PyAether** (158 PyAether tasks), a 157-task SKILL slice, and a 134-task Tcl slice (Custom WaveView 77 + Innovus 57), for **449 tasks in total**. Main results are on the PyAether slice; the cross-language reproduction is in Section 6.2.4; the Ω and Φ axes are PyAether-only.

EDA-Eval-PyAether (the 158-task PyAether slice) was collected from four sources: API references (61.4%), documentation examples (27.2%), internal training materials (7.6%), and anonymized CAD practical cases (3.8%), each manually reviewed by EDA domain experts. Layout and Schematic operations constitute the majority (70.2%: Layout 64 / 40.5%, Schematic 47 / 29.7%; Design Management 24 / 15.2%; Others 23 / 14.6%).

Evaluation Protocol. Each task consists of a natural language prompt, a function signature, and assertion-based test code. A task passes if the generated code executes in the PyAether sandbox without errors and all assertions pass. The primary metric is Pass@1 [6]:

$$\text{Pass@1} = \frac{\#\text{ tasks with all assertions passed}}{\text{Total tasks}} \times 100\%$$

6 Experiments

6.1 Experimental Setup

6.1.1 Research Questions. The framework of Section 4 makes three testable claims, one per research question:

- **RQ1 (Ceiling: observation or prior?).** Is accuracy bounded by Ω's fidelity or by $\mathcal{K}$'s strength? S1 varies Ω (LLM fixed); S3 varies $\mathcal{K}$ (harness fixed).
- **RQ2 (Which components are necessary, and interchangeable?).** Does bridging the gap require observation, a causal effector, and continuous Φ? S2 varies the effector and the belief-update budget.
- **RQ3 (Domain class or toolchain?).** Does the same structure reproduce on a structurally different toolchain? S1–S2 are re-run on the SKILL and Tcl slices of **EDA-Scripting-Bench**.

RQ1 and RQ2 test the structural claims; RQ3 tests whether they follow from the domain's structure rather than from one toolchain.

6.1.2 Benchmark and Evaluation Protocol. We evaluate ZhuLong on **EDA-Eval-PyAether** (158 tasks) using **Pass@1** as the primary metric. Each task runs in a single trace with per-execution timeout 2500 s; the agent may iterate within a trace via Φ but is never restarted. A file-based `PreToolUse` hook enforces a tool whitelist, blocks shell/code-search, and confines file access to a fresh empty workspace (identical across all arms). Each configuration is evaluated over five independent runs with reshuffled task order; accuracy is mean±std. Generation-stage failures (no executable code) are reported separately from execution-stage failures. Runs with infrastructure faults are discarded and re-run; the full-system anchor must reach ≥75% Pass@1 as a health floor. All figures are on the PyAether slice; cross-language results in Section 6.2.4.

6.1.3 Baselines and ZhuLong Variants. To isolate each component's contribution, we use four configurations: **Baseline 1 (Pure LLM)** (no retrieval, no sandbox), **Baseline 2 (RAG)** (retrieval only), **ZhuLong w/o Retrieval** (sandbox only), and **ZhuLong (Full)** (both). Retrieval and documentation form Ω; the sandbox is the effector. Each capability is toggled by a tool-registration switch—a single disabled-list in the server environment—so a disabled tool vanishes from the agent's tool set while every enabled tool keeps a byte-identical implementation across all arms. The sandbox is removed by unregistering `run_code`; retrieval by unregistering the three retrieval tools; the anti-leak hook stays loaded in every arm.

6.1.4 LLM Backbones. All main experiments use DeepSeek-V4-Pro (FP4) as the primary backbone; GLM-5.2, DeepSeek-V4-Flash, Kimi-K2.6, and Doubao-Seed-2.0-Pro are additionally evaluated (Section 6.2.3).

6.2 Results and Ablations

Table 1 gives the progressive build-up and component ablation as an overview; the studies below change one component at a time, with the LLM frozen unless stated otherwise.

Table 1: Ablation study results

Configuration	Pass@1 (%)	Δ
Progressive build-up (vs. previous row)		
Baseline 1: Pure LLM	10.5 ± 1.9	—
Baseline 2: RAG	72.2 ± 1.6	+61.7 pp
ZhuLong (Full)	**81.4 ± 1.5**	**+9.2 pp**
Component ablation (vs. ZhuLong Full)		
ZhuLong w/o Sandbox	72.2 ± 1.6	−9.2 pp***
ZhuLong w/o Self-Expl.	77.3 ± 2.9	−4.1 pp*
ZhuLong w/o Retrieval	81.0 ± 4.5	−0.4 pp^{ns}

McNemar + paired bootstrap (10 000 resamples, 95% CI) on 790 paired observations per arm: ***p¡0.001, *p¡0.05, ns = not significant. w/o Sandbox: $p<10^{-8}$, CI $[-13.1, -5.4]$. w/o Self-Expl.: $p<0.05$, CI $[-7.0, -1.2]$. w/o Retrieval: $p=0.81$ (ns).

6.2.1 S1: Observation fidelity (RQ1). Principle 2 predicts that with the model fixed, accuracy is bounded by Ω's fidelity. We vary Ω along two axes, holding the LLM fixed.

Documentation axis. Table 2 changes the documentation channel together with the effector's presence: (N) no observation; (L) a low-fidelity name-only index *with* the execution loop; (H) a high-fidelity index (adding descriptions, BM25, abbreviations, synonyms) used *without* the effector as the RAG baseline.

The documentation axis is non-monotone: (L) at 82.8% exceeds (H) at 72.2%—but this reflects the effector, not the

Table 2: Documentation fidelity with and without the effector (LLM and loop fixed; (L) includes the effector, (H) is the retrieval-only RAG baseline without it)

Rung	Configuration	Pass@1 (%)
(N) No observation	Baseline 1 (Pure LLM)	10.5 ± 1.9
(L) Low fidelity	name-only index	82.8 ± 1.9
(H) High fidelity	+ description, BM25, abbrev., synonyms	72.2 ± 1.6

index. (L) couples a coarse index with the full execution loop; (H) is retrieval-only. A coarse index *with* execution outperforms a rich index used only for static retrieval.

Read-back axis. Three closed-loop variants differ only in how much live state execution returns: (N′) prior only; (B) binary pass/fail; (F) full observation (stdout, exception traces, per-assertion outcomes, queried state—the default). The three rungs compress the *same* executed state to different bandwidths. Principle 2 predicts a monotone gain (N′)<(B)<(F). Table 3 reports Pass@1 together with convergence cost.

Table 3: Read-back axis: accuracy and convergence cost (LLM and effector fixed). (N′) cost is not recoverable (traces discarded in a storage migration); (F)/(B) cost columns are 5-run means ± sample std.

Read-back	Pass@1 (%)	Mean `run_code`	Wall (h)	Timeout
(N′) None (prior only)	73.3 ± 2.2	not recoverable†		
(B) Binary pass/fail	78.1 ± 1.3	22.3 ± 3.3	3.73 ± 0.72	7.6 ± 3.1
(F) Full observation	81.4 ± 1.5	12.1	2.22	3.4*

* mean of 5 runs (range 1–6). † (N′) traces discarded; cost not imputed.

The results recover the predicted monotone ordering (N′)<(B)<(F). N′ → B (+4.8 pp) is the gain from learning whether a run passes at all; B → F (+3.3 pp) is the further gain from seeing the concrete error.[1] Convergence cost moves oppositely: dropping (F)→(B) inflates mean `run_code` calls ∼1.8× (12.1→22.3), wall-clock ∼1.7× (2.22→3.73 h), and timed-out tasks ∼2.2× (3.4→7.6)—the "fidelity down → convergence cost up" trade-off Principle 2 implies.

6.2.2 S2: The closed loop (RQ2). Principle 3 requires actions to reach the design through a causal interface, and Φ to run continuously.

Effector. Removing the sandbox drops Pass@1 from 81.4% to 72.2% (−9.2 pp; McNemar $p<10^{-8}$, 95% CI [−13.1, −5.4] pp); adding it to the RAG baseline raises it by the same amount. Without an effector, b_t receives no evidence about the live state and the agent is reduced to a prior-only predictor. Execution returns concrete error messages that guide corrections, transforming one-shot generation into iterative search [10, 12].

[1] An earlier legacy probe of the (B) arm showed an N′-over-B inversion (66.2±16.1%, one anomalous run at 43.0%); the fresh 5-run re-measurement collapses the spread to ±1.3 and restores the predicted ordering.

Belief update. ZhuLong is autonomous, so we do not shorten the loop or force early submission. Instead we bound how often the loop *may execute*—the number of `run_code` calls per trace—which determines how much state evidence b_t receives. Once the budget is spent, the call is denied at the tool boundary; the agent keeps running under its own control until it submits. Table 4 also includes a *lagged* channel: a single-slot (L=1) buffer whose read-back returns the state from the *previous* execution, leaving the agent permanently one action behind the design. Principle 3 predicts accuracy rises with the execution budget but saturates once updates outpace the design's dynamics.

Table 4: Bounding the evidence available to Φ (LLM, Ω, effector fixed). k = max `run_code` calls per trace; "lagged" = read-back returns the *previous* execution's state.

Update budget	Pass@1 (%)	Conv. (%)	Gen-fail
Φ k=1	63.8 ± 3.2***	76.3	34.6
Φ k=3	69.1 ± 2.5***	80.4	27.6
Φ k=10	77.7 ± 2.2**	90.4	17.4
Φ lagged	73.0 ± 3.2***	97.5	9.0
Φ unbounded	81.4 ± 1.5	99.1	10.4

5-run means; unbounded = full-system anchor. *Gen-fail* = per-run count of tasks producing no executable code. ***p¡0.001, **p¡0.01 vs. unbounded (McNemar + bootstrap).

Reading Table 4 with the *Mean gen-fail* column reveals the mechanism: as k shrinks, tasks producing no executable code rise* (17.4 at k=10, 27.6 at k=3, 34.6 at k=1), driving Pass@1 downward. The budget is exhausted at the tool boundary, but the agent keeps running; with no further `run_code` calls it cannot finish or validate its generation—the $k\downarrow\rightarrow$ gen-fail$\uparrow\rightarrow$ Pass@1$\downarrow$ signature of Φ being bounded, consistent with Principle 3.

6.2.3 S3: The prior axis, the contrast arm of RQ1. Principle 1 separates knowledge from belief: the frozen prior should bound what a harness can achieve without substituting for observation. Table 5 changes only $\mathcal{K}$—the LLM—with the harness fixed. We use DeepSeek-V4-Pro (FP4) as the primary backbone for its cost–capability trade-off; GLM-5.2 attains marginally higher Pass@1 (83.3%) but overlaps within one standard deviation. Read against Principle 2, this spread is what $\mathcal{K}$ can move—smaller than the observation-channel effects in S1–S2, confirming that the prior does not substitute for observation.

6.2.4 RQ3: re-measuring the structure per toolchain. The three slices of **EDA-Scripting-Bench** share one task schema, one assertion criterion, and one harness, so S1–S2 can be re-run slice by slice with nothing changed but the toolchain. Table 6 reports the component ablation on SKILL (157) and Tcl (134, aggregating WaveView and Innovus), alongside the PyAether anchor; the *Overall* column pools all 449 tasks. The comparison of interest is the *within-language* ordering.

Table 5: Impact of LLM backbone (ZhuLong)

LLM Backbone	Pass@1 (%)	Δ
DeepSeek-V4-Pro	81.4 ± 1.5	—
GLM-5.2	83.3 ± 3.1	+1.9 pp^{ns}
DeepSeek-V4-Flash†	60.1 ± 5.5	-21.3 pp***
Kimi-K2.6	77.0 ± 1.6	-4.4 pp**
Doubao-Seed-2.0-Pro	63.8 ± 1.4	-17.6 pp***

Significance vs. DeepSeek-V4-Pro (McNemar, paired bootstrap 95% CI): ***p¡0.001, **p¡0.01, ns = not significant. Flash†: retest 5-run under reduced gateway parallelism ($-p$ 4) and enlarged token budget; the initial 5-run (11.9%, depressed by an infrastructure confound) is superseded.

Table 6: Cross-language reproduction (RQ3). Pass@1 (%), 5-run mean±std on PyAether/SKILL; Tcl aggregates Custom WaveView (77) + Innovus (57) into a 134-task column; Overall pools all 449 tasks. The full system is the top configuration on every language; the prior-only baseline is the bottom on every language.

Configuration	PyAether	SKILL	Tcl	Overall
Pure LLM	10.5 ± 1.9	33.2 ± 2.3	10.2	18.3
RAG (retrieval only)	72.2 ± 1.6	33.9 ± 10.6	84.6	62.5
ZhuLong w/o Retrieval	81.0 ± 4.5	39.0 ± 12.6	73.2	64.0
ZhuLong (Full)	**81.4 ± 1.5**	**60.3 ± 6.7**	**87.2**	**75.8**

The qualitative structure reproduces across all three languages: the full system is the *top* and the prior-only baseline the *bottom* on every language—the dominance of observation-plus-execution is domain-structural, not toolchain-specific (RQ3). The intermediate ordering is less stable (sandbox-only beats RAG on PyAether, the reverse on Tcl, close on SKILL), reflecting that different toolchains expose different amounts of usable documentation relative to executable feedback. The Ω and Φ axes are reported on PyAether only.

6.3 Error Analysis

To characterize failure modes, we inspected a representative trace from the full-system 5-run set—the run whose Pass@1 (81.6%, 129/158) is closest to the 5-run mean (81.4±1.5%). The 29 failures comprise 17 generation-stage failures (no executable code produced) and 12 execution-stage failures, classified by the L1–L4 hierarchy of Section 4.

The dominant execution-stage failure mode is **L4** (8/12, 67%): situated truthfulness violations that no offline knowledge can prevent—the code is syntactically correct (L1), uses documented APIs with plausible arguments (L2–L3), yet fails because it violates an *undocumented* binding contract, depends on an internal selection state, or assumes a runtime context that does not hold. For example, Task 010 (`add_instance_to_schematic`) executes and passes the agent's own assertions, yet the official harness rejects it: `target_cv must be an opened cellview object`—an L4 property of the live design state, not of the API signature. Tasks 063, 074, 146, 155 follow analogous patterns (hidden type names, attributes absent in the test path, GUI-dependent return values). The remaining execution-stage failures are **L3** (4/12, 33%): valid APIs composed with incorrect logic (Task 052 substitutes a custom validator for the mandated call; Task 058 uses substring matching instead of equality). **L1/L2 = 0%**.

The 17 generation-stage failures are timeout (4), wrong save path (10, saving to `ws/` instead of `ws/pyAether_code/`—a procedural error amenable to harness-level mitigation), and no code produced (3).

The L1–L4 classification makes the grounding-gap framework concrete: 67% of execution-stage failures are L4, invisible until the agent acts on the live design and reads back the result. This is why the execution loop is decisive: the sandbox is the only channel through which the agent can observe hidden type contracts, internal selection states, and runtime-context dependencies. These findings motivate type-aware API documentation and case-level knowledge distillation.

7 Discussion

The error analysis (Section 6.3) shows that 67% of execution-stage failures are L4 (situated truthfulness) violations caused by undocumented API contracts and hidden type signatures, motivating type-aware retrieval and case-level knowledge distillation. More broadly, the dominance of sandbox execution (+9.2 pp) over static knowledge elevates execution grounding from an empirical observation to a structural principle: the grounding gap is bounded by observation fidelity, so execution channels—not larger priors—are the binding constraint for domains with long-tailed, under-documented APIs. This paradigm may generalize to other domain-specific scripting tasks with similar characteristics.

Limitations

Several limitations apply. (1) Performance varies across base LLMs (60.1%–83.3%); the model remains a decisive factor. (2) The interactive bridge is single-threaded. (3) Main results are on the 158-task PyAether slice; the Ω and Φ axes are PyAether-only (cross-language component ablation in Section 6.2.4). (4) Benchmark size limits statistical power for small effects; we report McNemar/bootstrap for main contrasts but treat small gaps (e.g. −0.4 pp, p=0.81) as directional. (5) The primary backbone is FP4-quantized; quantization effect is not ablated. (6) Convergence-cost comparison is complete for (F) and (B) only; (N′) traces were discarded.

8 Conclusion and Future Work

ZhuLong is an execution-grounded LLM coding agent for PyAether, SKILL, and Tcl that closes the generate–execute–refine loop via sandbox feedback. Its theoretical grounding—EDA correctness is a situated (L4) property, and the grounding gap is bounded by observation fidelity, not model scale—explains why runtime observation is necessary. On **EDA-Eval-PyAether** (158 real-world tasks), the complete system

achieves 81.4% Pass@1 (±1.5), substantially outperforming static RAG (72.2%); ablation shows execution and retrieval are jointly necessary. This work establishes three foundations: (1) the grounding-gap theory, (2) the execution-grounded agent paradigm generalizable to other long-tail-API domains, and (3) **EDA-Scripting-Bench** (449 tasks across three toolchains) as evaluation infrastructure. The harness, task schema and protocol will be released.

Future work proceeds along three directions. First, extending all ablation axes to the full EDA-Scripting-Bench (PyAether + SKILL + Tcl, 449 tasks), together with extending the benchmark to systematically cover GUI-bridged tasks on open design sessions. Second, evolving the agent through self-improvement—learning from both successful and failed execution traces to refine its retrieval, code generation, and exploration strategies. Third, adapting the architecture to additional EDA platforms, including Synopsys tools with Tcl scripting, to broaden the scope of execution-grounded EDA agents.

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